IMPERIAL

ELEC40006-Electronics Engineering Design Project 1

Rover to the Moon!

Dr. Hakan Merdan

Rover to the Moon!

Aims of the Module

To bring together the knowledge and skills you have gained this academic year
Apply knowledge and skills to solve a practical problem
Acquire new skills that will help you on this and future projects:

- Project Management
- Engineering Design
- Documentation
- Group Work
- Problem Solving
- Professional competencies

Rover to the Moon!

Geological Survey!



to design, build and test a rover for geological surveying of the Moon's surface to identify rocks

Rover to the Moon!
The Challenge!

Find out each rocks'

Age

- Communicated with radio signals

Type

- Discovered with infrared pulses, ultrasound, and direction of magnetic field



EEELunar Rover

Specifications

Functional Requirements

1. Find out the age, type of each rock
2. Manoeuvre quickly and accurately around a test arena
3. Follow user commands from an app or webpage interface, and relay discoveries back to the user

Non-Functional Requirements

1. Lightweight
2. Robust and reliable
3. Cost effective
4. Easy to use

Full specifications can be found in the project brief

EEELunar Rover
Deliverables

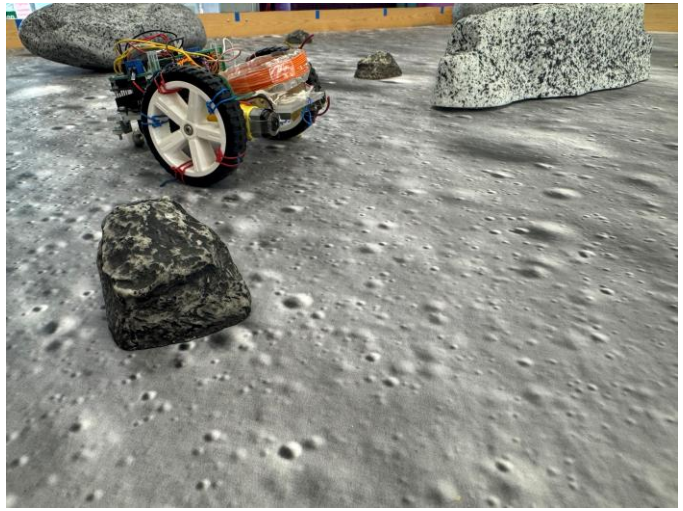
Assessment	Date	Content	Weighting
Interim Presentation	28 May	A presentation with Q&A to exhibit your design so far	20%
Reflections on professional practice	11 June	Reflection on your professional development and how it relates to the project.	0% (Must pass component)
Project report	11 June	A formal technical report detailing the design process and delivered technical solution	40%
Interview, Demo and Competition	16 June	Explain to the examiner how your EEELunarRover works and answer their questions. Then compete head-to-head with other groups to compete the challenge in the quickest time	40%

EEELunar Rover

Demo Arena/Imperial Lunar Surface







A number of rocks, obstacles
Available for testing in the lab.

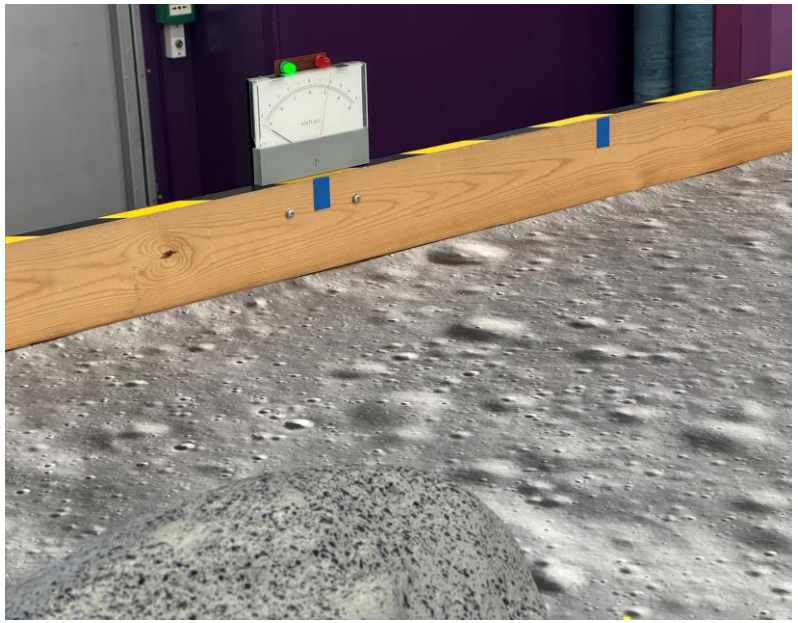
EEELunar Rover
Weight Sensitive Zone

Weight sensitive zone on the lunar surface can be considered as an extremely loose and fragile layer that heavy rovers will fail classifying the rock there.

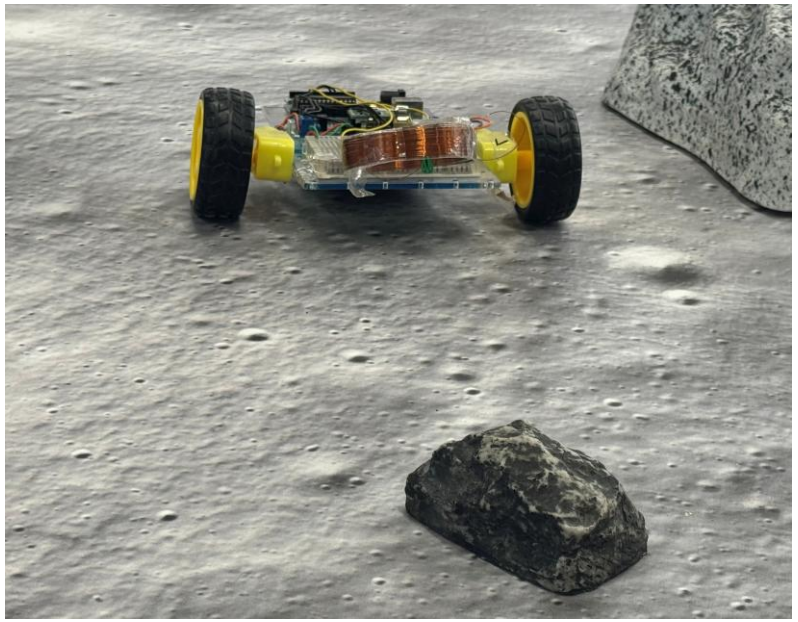
The sensitive zone will show if your rover is too heavy (about more than 750gr)

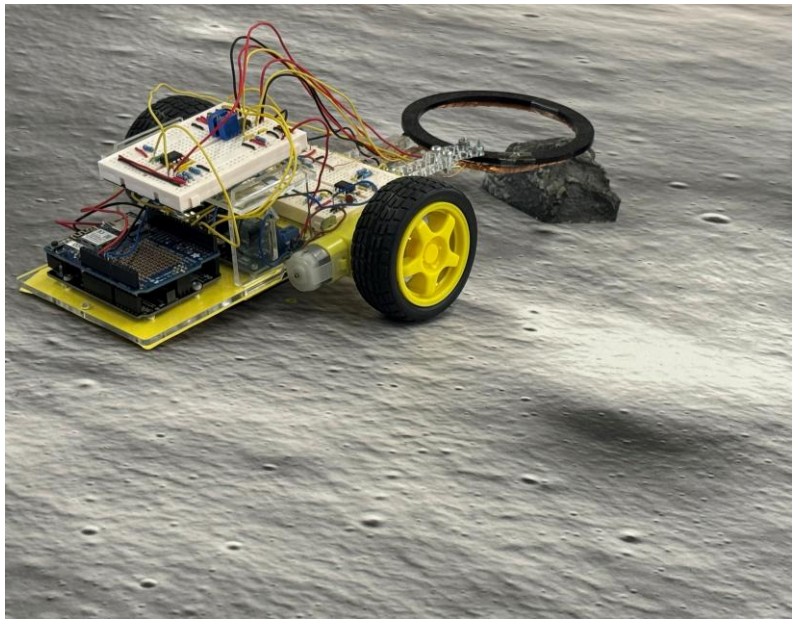
In the demo, the rock here will not be classified if your rover is too heavy.

EEELunar Rover must be lighter than 750gr to read accurate data from the rock in that sensitive zone.









What do you need to know?

EEELunar Rover

What do you need to know?

Project Brief:

Top-level requirements

Definitions of rock properties

List of deliverables and mark schemes

Technical hints and starter kit instructions

Autumn and Spring Content

Prototyping (labs)

Test and diagnosis (lab skills, ADC)

Resonant Circuits (ADC)

Amplification (ADC)

C++ programming (Programming)

Embedded programming (lab skills)

Digital data representation, state and logic (DECA)

EEELunar Rover

Self-guided learning

Research

Sensing

Circuit designs and algorithms

Mechanical design

Embedded Programming

Make connections between modules

Be willing to learn and experiment





Organisation

EEELunar Rover
Group Working

Pre-allocated groups of 6 students. They are shared in Teams channel under 'Shared' section.

- Overlap with lab/tutorial groups

One mark for the whole group

- Share the work fairly

Develop team management skills

- Share the burden of difficult challenges
- Illness, absence: reassign the work. Allow contingency

Adapt to the resources available

- Escalate persistent problems with fair workload

Deliverables contain marks for teamworking



EEELunar Rover
Working in the Lab

Year 1 and Year 2 are working on group projects this term
Space in the lab for around half of these cohorts

- Make good use of morning opening hours
- Bench booking may be introduced if demand is high
- You may work outside the lab, but
  - Don’t do hardware work in the central library
  - Don’t remove the equipment from the lab. You can borrow a Picoscope from Stores
  - Take extra care with your work, especially **BATTERIES**.

EEELunar Rover

Safety Risk Assessment

BATTERIES

DO NOT RECHARGE THEM.

DO NOT TRY TO RECHARGE THEM.

DO NOT RECHARGE THEM UNDER ANY CIRCUMSTANCES.

BE VERY CAREFUL WITH THEM, DO NOT DROP THEM, PUNCH THEM OR SHORT THEM ACCIDENTALLY.

DO NOT PUT THEM NEXT TO CONDUCTING MATERIALS OR COVER THEM WITH. THAT MIGHT MAKE THE TERMINALS SHORTED AND THE BATTERY MIGHT EXPLODE.

DO READ TECH AND SAFETY DATASHEET BEFORE USING THE BATTERIES

DO NOT USE THEM OUTSIDE THE LABS.

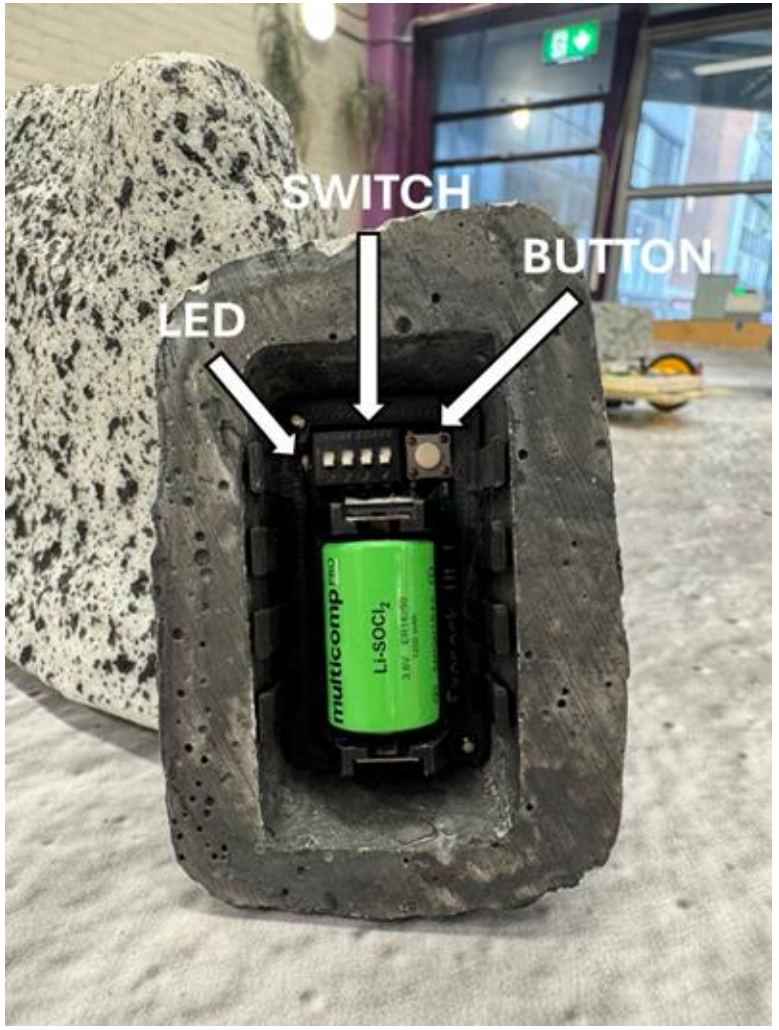
CHECK TERMINALS AND POLARITY BEFORE INSERTING THEM.

ONLY USE THE BATTERIES PROVIDED.

EEELunar Rover

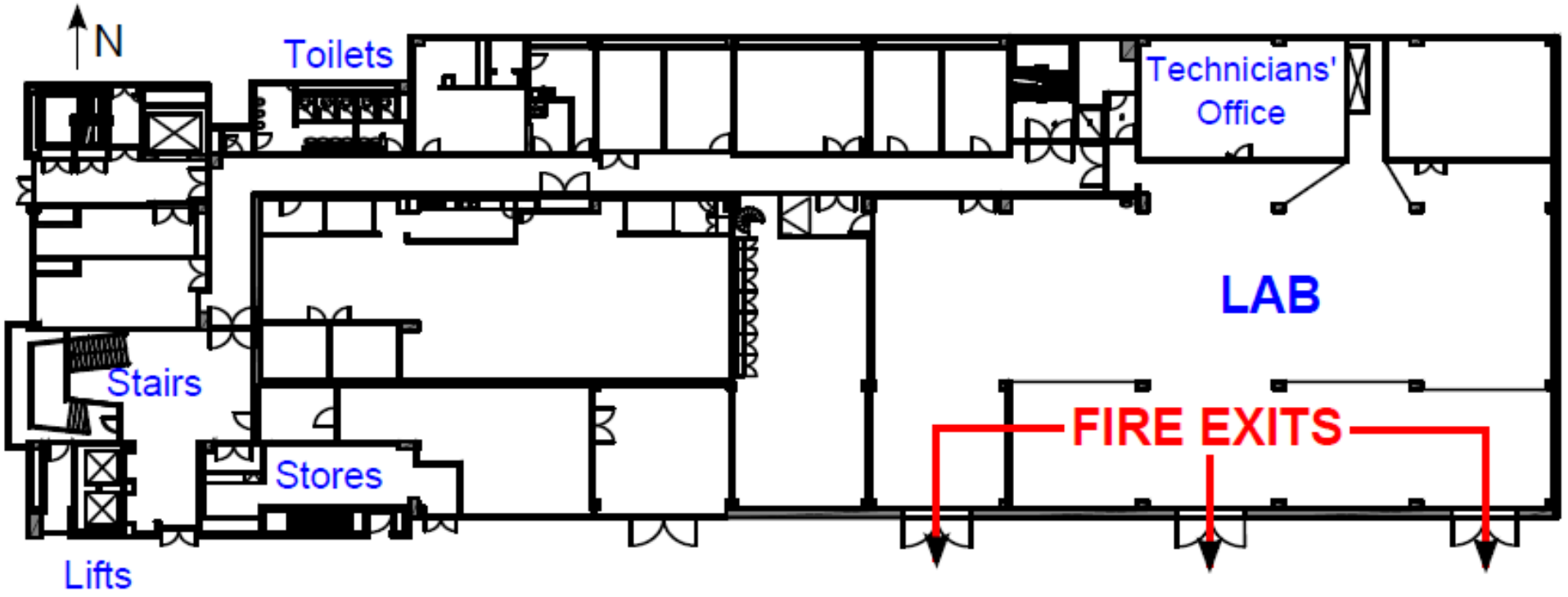
Rock Simulator Configuration

The rock is configured by setting a group of DIP switches on its internal PCB. The switch, button, and LED are accessible without removing the PCB from the rock simulator.



EEELunar Rover
Safety Risk Assessment

FIRE EXITS



EEELunar Rover
Getting Help

[Consultation Session Booking]

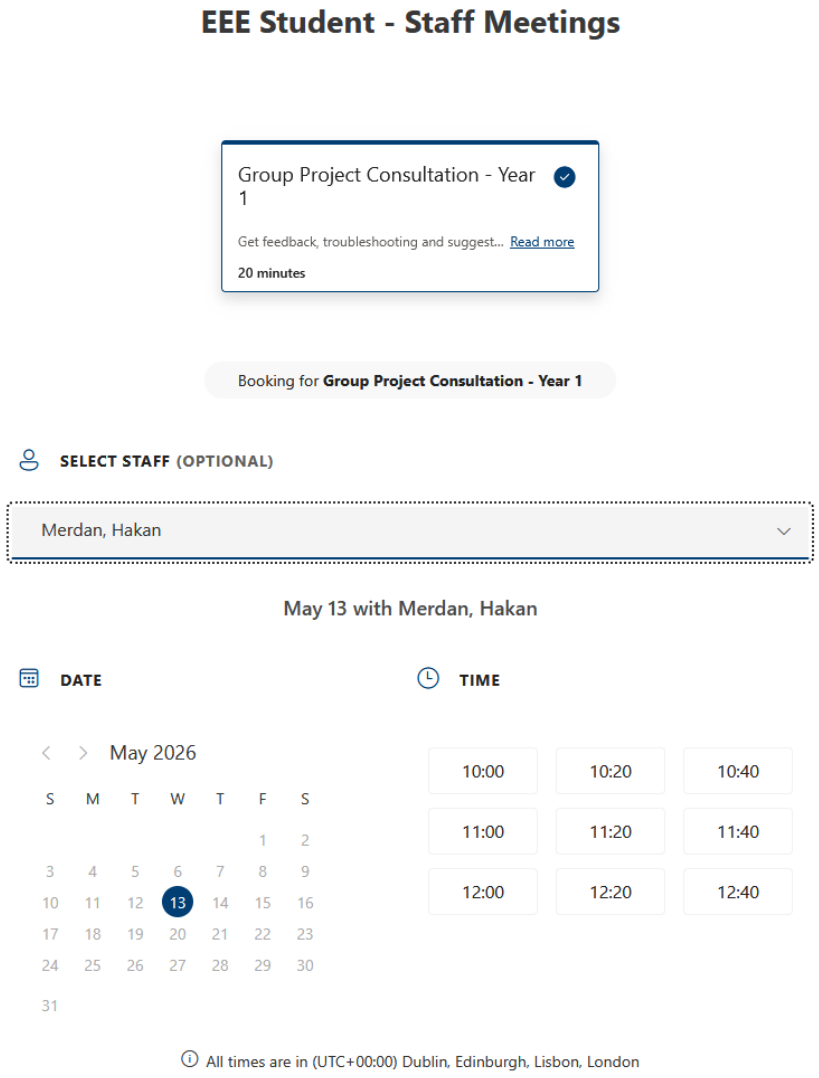
Max. one booking per group per week

- Unless there is availability on the current day

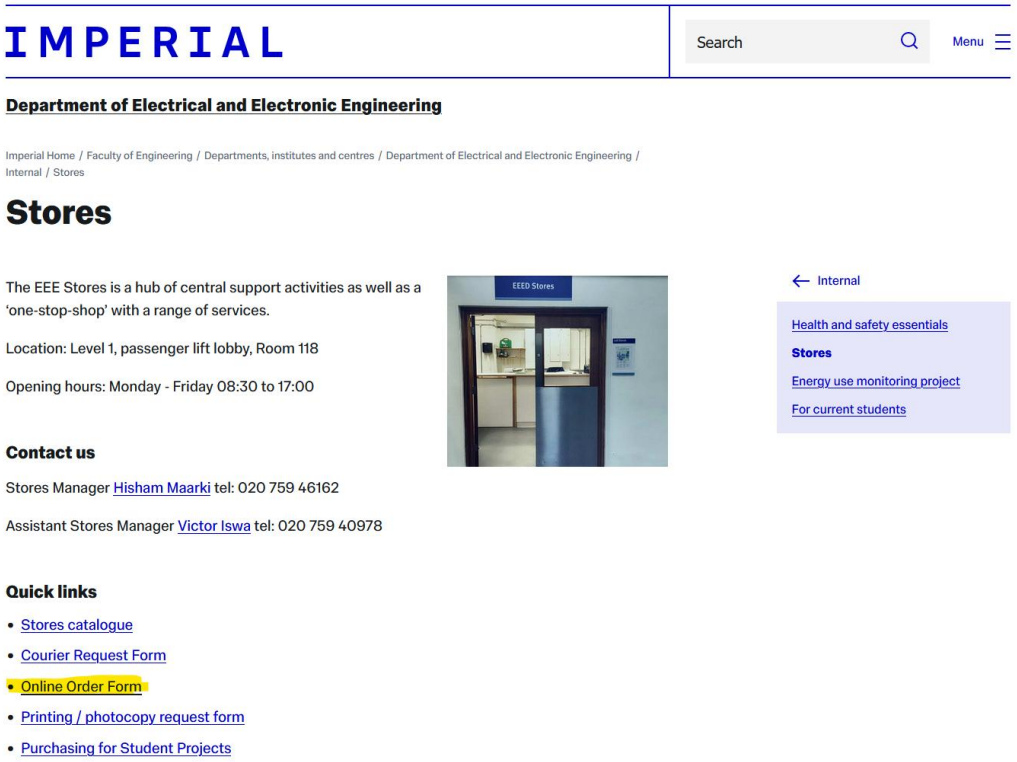
You can book up to 1 week in advance

Cancel your booking if it is no longer needed

Staff timetable and expertise will be also published on Teams.



EEELunar Rover
Ordering Parts



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Search Menu

Department of Electrical and Electronic Engineering

Imperial Home / Faculty of Engineering / Departments, institutes and centres / Department of Electrical and Electronic Engineering / Internal / Stores

Stores

The EEE Stores is a hub of central support activities as well as a 'one-stop-shop' with a range of services.

Location: Level 1, passenger lift lobby, Room 118

Opening hours: Monday - Friday 08:30 to 17:00

EEED Stores

← Internal

Health and safety essentials
Stores
Energy use monitoring project
For current students

Contact us

Stores Manager Hisham Maarki tel: 020 759 46162

Assistant Stores Manager Victor Iswa tel: 020 759 40978

Quick links

- Stores catalogue
- Courier Request Form
- Online Order Form
- Printing / photocopy request form
- Purchasing for Student Projects

https://www.imperial.ac.uk/electrical-engineering/internal/stores/

You can order components for your rover

Follow the department's website to familiarise with order via Stores

- Contains instructions and links to suppliers
- Will be linked on Teams.
- Each group must nominate an approver to approve orders
- Receive notification when the order is ready to collect from Stores
- The approver will follow the component orders behalf of the group via Stores website.

Check details carefully before ordering

- Check Datasheet
- In stock?
- Pack size / minimum quantity
- Watch out for surface mount devices

Budget allowance is £60 for each group.

Getting Started

Meet your group

Check your group allocation on Teams
Most group members will be familiar from tutorials and labs
Set up a group chat
Some people have been moved to balance groups – please make sure everyone is included from the start



EEELunar Rover
Read the documents (Starting from the next week)

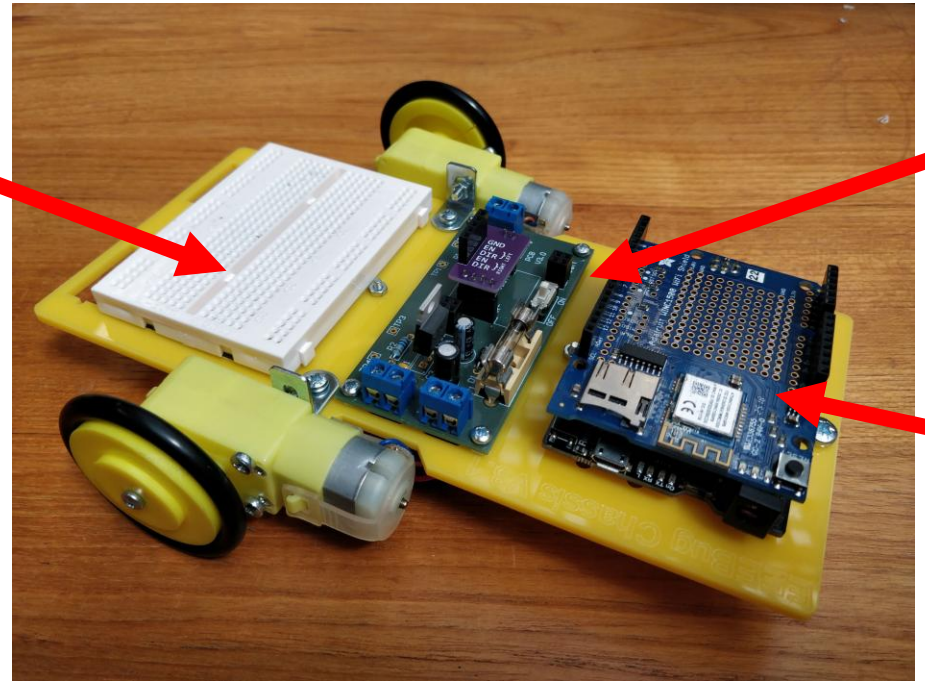
GitHub documentation

https://github.com/hakanmerdan/EEELunarRover2526

- Project Brief
- Technical hints
- Starter Code
- Design Files
- EEEBug expansion kit ready for collection on Wednesday
  - Includes a ‘rock simulator’

Collect your expansion kit

Add sensors and circuits



Motor Driver Module

Adafruit Metro M0 with WiFi shield

Rover Platform



Rock simulator

Available from lab technicians from Wednesday 13th May

Listen to the advice from last year's students

If you started the project today knowing what you do now, what would you do differently?

Invest more time in the initial planning phase. I have realised the importance of setting clear objectives, defining project scope, and establishing a realistic timeline ...to ensure a more focused and well-defined project direction.

Assign team members according to skill instead of aspirations

In this project, team success is the ultimate goal, and I believe that hands-on experience is the key form of learning that will help us grow as engineers in the best way possible

Accept losses and go for simpler solutions


Focus more on the report as I feel as we put almost all of our efforts and attention on the actual product, while not creating a report as good as the final rover
I would start the design and setup earlier and open more time for testing
Escalate team member issues if we cannot resolve ourselves
Leave plenty of time for integration
Select a project manager in the group

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Thank you!
Any Questions?

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Professional Skills

Design Project 1 2026

Prabhav Reddy
08/05/2026

Agenda

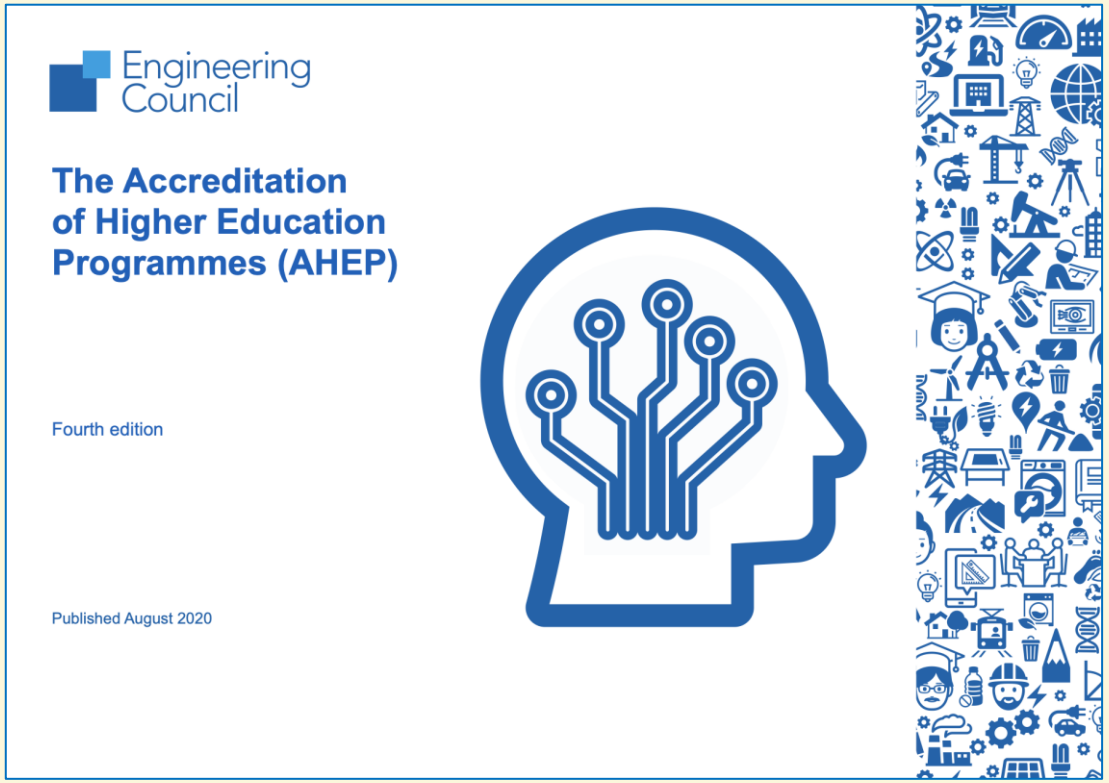
1. Professional Competencies
2. Assessments
3. Next Steps

Agenda

1. **Professional Competencies**
2. Assessments
3. Next Steps

Professional Competencies
Accreditation

- Our courses are accredited by the Institution of Engineering and Technology
- UK Standard for Professional Engineering Competence
- AHEP4 has five areas of interest:
  - Science and Mathematics
  - Engineering Analysis
  - Design and Innovation
  - The Engineer and Society
  - Engineering Practice



https://www.engc.org.uk/media/3410/ahep-fourth-edition.pdf

Five Areas of Professional Competencies

Area	Description
Science and Mathematics	The study of engineering requires a substantial grounding in engineering principles, science and mathematics commensurate with the level of study
Engineering Analysis	Engineering analysis involves the application of engineering concepts and tools to analyse, model and solve problems. At higher levels of study engineers will work with information that may be uncertain or incomplete.
Design and Innovation	Design is the creation and development of an economically viable product, process or system to meet a defined need. It involves significant technical and intellectual challenges commensurate with the level of study.
The Engineer and society	Engineering activity can have a significant societal impact and engineers must operate in a responsible and ethical manner, recognise the importance of diversity, and help ensure that the benefits of innovation and progress are shared equitably and do not compromise the natural environment or deplete natural resources to the detriment of future generations.
Engineering practice	The practical application of engineering concepts and tools, engineering and project management, teamwork and communication skills. Engineers also require a sound grasp of the commercial context of their work, specifically the ways an organisation creates, delivers and captures value in economic, social, cultural or other contexts.

Professional Competencies

AHEP4 Learning Outcomes

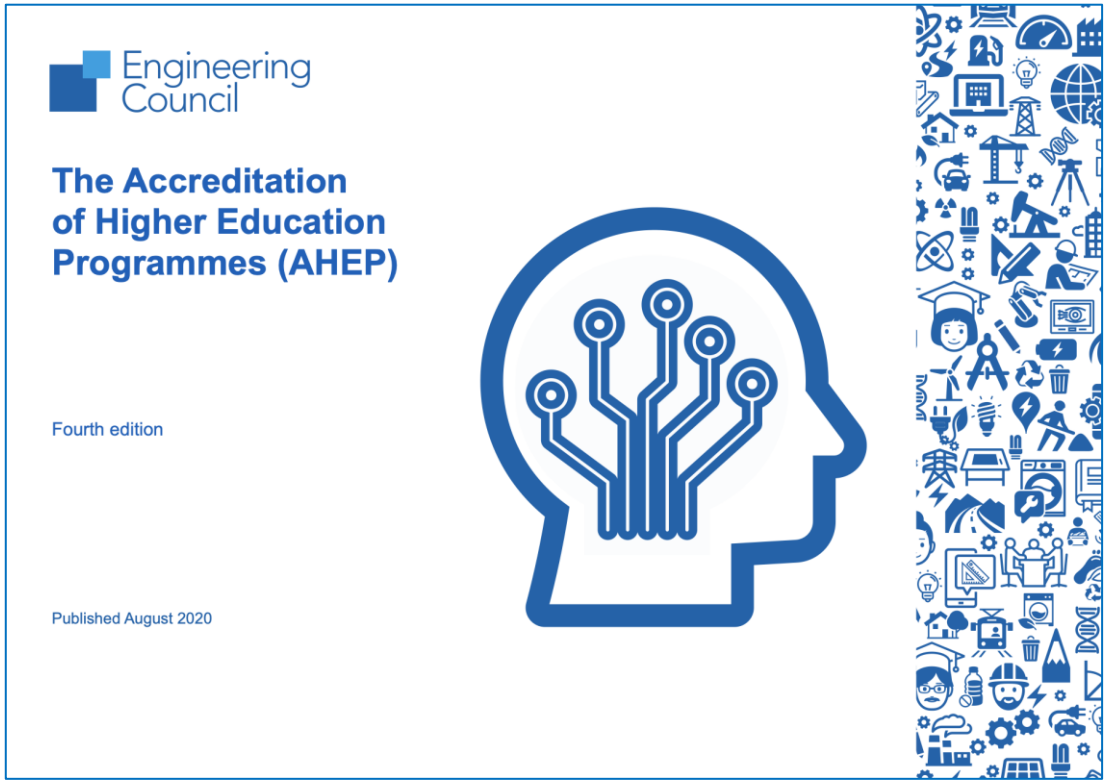
Science and Maths	M1	Science, mathematics and engineering principles	Apply a comprehensive knowledge of mathematics, statistics, natural science and engineering principles to the solution of complex problems. Much of the knowledge will be at the forefront of the particular subject of study and informed by a critical awareness of new developments and the wider context of engineering
Engineering Analysis	M2	Problem analysis	Formulate and analyse complex problems to reach substantiated conclusions. This will involve evaluating available data using first principles of mathematics, statistics, natural science and engineering principles, and using engineering judgment to work with information that may be uncertain or incomplete, discussing the limitations of the techniques employed
	M3	Analytical tools and techniques	Select and apply appropriate computational and analytical techniques to model complex problems, discussing the limitations of the techniques employed
	M4	Technical literature	Select and critically evaluate technical literature and other sources of information to solve complex problems
Design and Innovation	M5	Design	Design solutions for complex problems that evidence some originality and meet a combination of societal, user, business and customer needs as appropriate. This will involve consideration of applicable health & safety, diversity, inclusion, cultural, societal, environmental and commercial matters, codes of practice and industry standards
	M6	Integrated/systems approach	Apply an integrated or systems approach to the solution of complex problems
The Engineer and Society	M7	Sustainability	Evaluate the environmental and societal impact of solutions to complex problems (to include the entire life-cycle of a product or process) and minimise adverse impacts
	M8	Ethics	Identify and analyse ethical concerns and make reasoned ethical choices informed by professional codes of conduct
	M9	Risk	Use a risk management process to identify, evaluate and mitigate risks (the effects of uncertainty) associated with a particular project or activity
	M10	Security	Adopt a holistic and proportionate approach to the mitigation of security risks
	M11	**Equality, diversity and inclusion**	**Adopt an inclusive approach to engineering practice and recognise the responsibilities, benefits and importance of supporting equality, diversity and inclusion**
Engineering Practice	M12	Practical and workshop skills	Use practical laboratory and workshop skills to investigate complex problems
	M13	Materials, equipment, technologies and processes	Select and apply appropriate materials, equipment, engineering technologies and processes, recognising their limitations
	M14	Quality management	Discuss the role of quality management systems and continuous improvement in the context of complex problems
	M15	Engineering and project management	Apply knowledge of engineering management principles, commercial context, project and change management, and relevant legal matters including intellectual property rights
	M16	**Teamwork**	**Function effectively as an individual, and as a member or leader of a team. Evaluate effectiveness of own and team performance**
	M17	**Communication**	**Communicate effectively on complex engineering matters with technical and non-technical audiences, evaluating the effectiveness of the methods used**
	M18	**Lifelong learning**	**Plan and record self-learning and development as the foundation for lifelong learning/CPD**

Professional Competencies
Accreditation

Home Exercise:

Look at the AHEP4 criteria.

- Which of these criteria do you think we have taught and assessed?
- Which of these criteria do you think the group project will help you develop?
- How would you demonstrate that you have achieved these criteria?

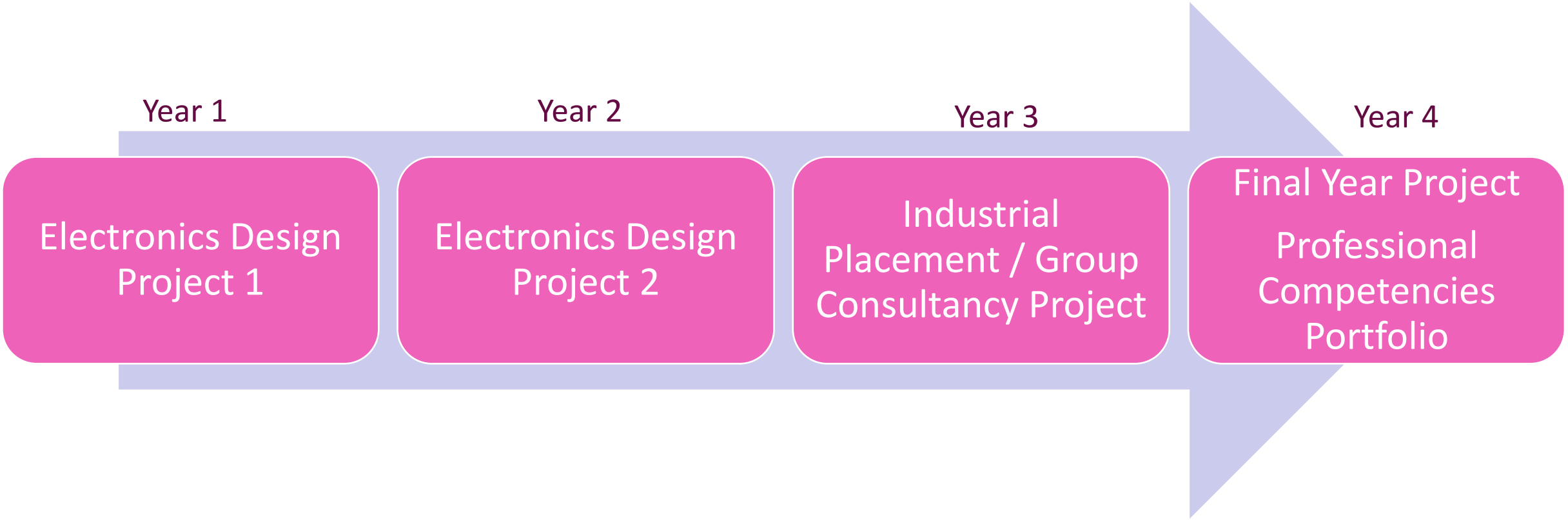


https://www.engc.org.uk/media/3410/ahep-fourth-edition.pdf



"Do we have to equip ourselves with tools for the future? Can't we just learn stuff instead?"

Professional Competencies In Your Course



Yr 1

- Teamwork
- Professional Competencies
- Technical Communication: reports/ presentations/ demonstrations

Yr 2

- Self-Learning
- Security
- Technical Communication: reports/ presentations/ demonstrations

Yr 3

- Group Design Project
- Industrial Placement

Yr 4

- Final Year Project
  - Research competencies
  - Technical comm: report writing /poster/ presentations
- Professional competencies portfolio
  - Critical thinking
  - Ethics and sustainability
  - EDI

Professional Portfolio - MEng Year 4 Term 1

Imperial College London

First Name	Surname	CID	IET Membership Y/N (Optional)

Student Report - reflecting on learning towards becoming a professional electrical engineer

Please outline in ~250 words your professional engineering development since starting your degree in the EEE Dept. Look at the items in column "EC UK Spec Code" that are considered relevant by the UK Engineering Council to reflect on what you have learned, what you have done, what would you do differently, what additional support you might need towards becoming a professional engineer. This is an evidence-based exercise, thus if you claim a 3 - skilled level you must look for evidence in what you have done to justify this claim.

EC UK Spec Code	*Claimed by Student*	*Assessed by Mentor*	*Agreed by Student and Mentor*
A1 - Use of Knowledge & Understanding	0 Non-Demonstrated	0 Non-Demonstrated	0 Non-Demonstrated
A2 - Application of Technical Skills	0 Non-Demonstrated	0 Non-Demonstrated	0 Non-Demonstrated
B1 - Problem Identification	0 Non-Demonstrated	0	0
B2 - Resource Management	0 Non-Demonstrated	0	0
B3 - Techniques and Procedures	0	0	0
C1 - Personal Responsibility	0	0	0
C2 - Leadership	0	0	0
C3 - Teamwork	0	0	0
C4 - Continuous Improvement	0	0	0
D1 - Communication Skills	0	0	0
D2 - Interpersonal Skills	0	0	0
D3 - Networking	0	0	0
E1 - Professional Conduct	0	0	0
E2 - Commitment to Continuous Learning	0	0	0
E3 - Understanding Professional Standards	0	0	0
E4 - Social Responsibility	0	0	0
E5 - Commitment to Diversity and Inclusion	0	0	0

Mentor Comments

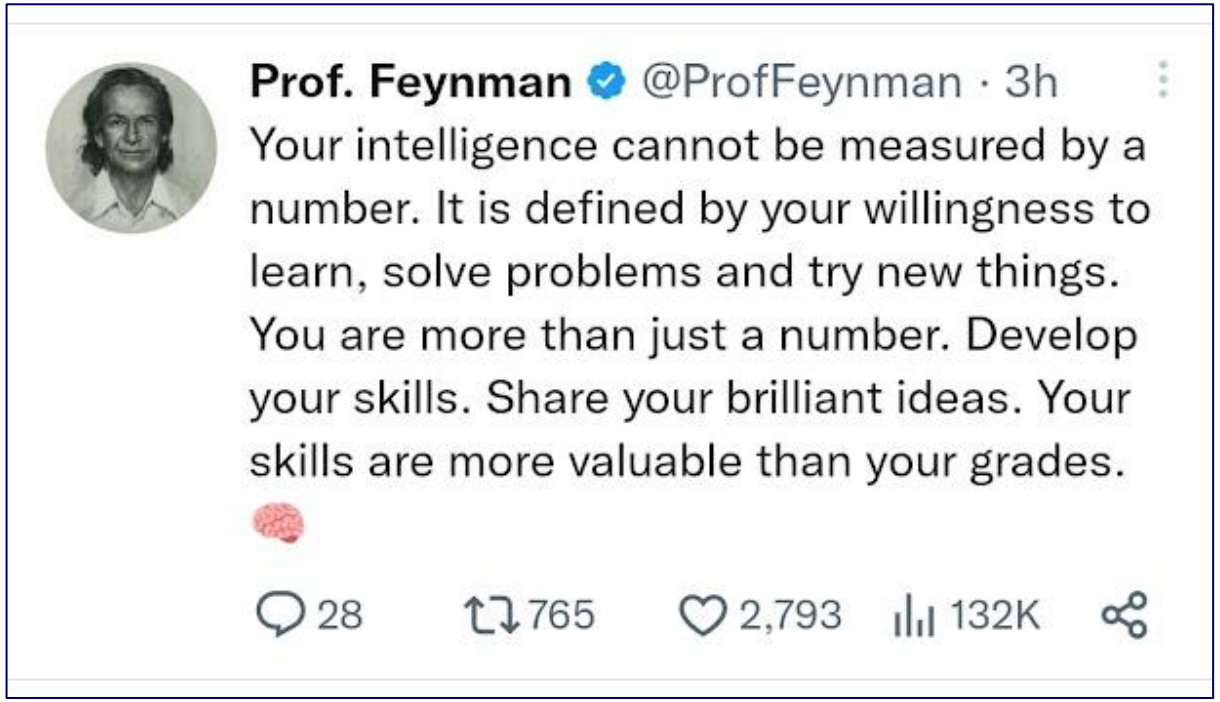
In this first mentoring session, the mentor's task is to guide the student into analysing which modules the student has taken delivers one or more of the EC UK Spec Code requirements and to what level. Group projects, internships, volunteering inside and outside the curriculum adds additional experience that will allow multiple Specs to be covered. The aim is not to tick all the boxes at the highest level (that cannot be achieved over 4 years of full time studies) but to reflect how the curriculum and additional experience are contributing towards becoming a professional engineer. Talk about how to look at achievements in an engineering context and to think about what modules/experience are required to fulfil more of the Spec requirements before graduating.

Electronically Signed by Student:	
Electronically Signed by Mentor:	
Date:	

REFERENCES:

Competence Levels	
0 Non-Demonstrated	Likely to be at this level at the start of studies.
1 Aware	Performs basic routine tasks, can recognise fundamental principles.
2 Familiar	Demonstrate familiary with engineering tasks and can apply knowledge in certain situations
3 Skilled	Demonstrates problem-solving abilities and can adapt new skills to new challenges.
4 Expert	Leading projects or initiatives, capable of developing others within the activity, mentoring peers.

Professional Portfolio



More information:
https://en.wikipedia.org/wiki/Richard_Feynman

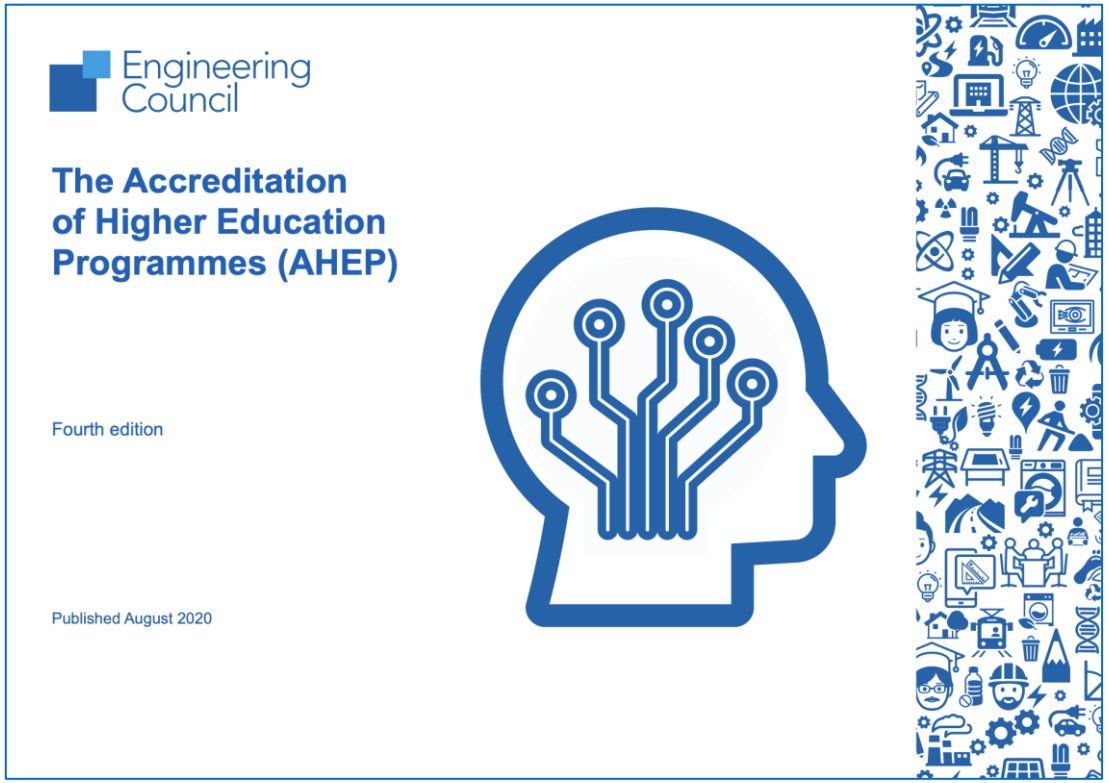
Agenda

1 Professional Competencies
2 **Assessments**
3 Next Steps

Professional Competencies

The competencies that we want to focus on are

- 11. EDI
- 16. Teamwork
- 17. Communication
- 18. Self-Learning



https://www.engc.org.uk/media/3410/ahep-fourth-edition.pdf

Teamwork and EDI

<table>
<tr><th>How is it assessed?</th><th>What is assessed?</th></tr>
<tr><td>Report</td><td rowspan="3">Individual contribution

Functioning of the team</td></tr>
<tr><td>Presentation</td></tr>
<tr><td>Demonstration</td></tr>
<tr><td>Individual Reflection (along with your report) – 300 words</td><td>Reflect on
Your contribution to the work
How the team worked together?
How a diversity of people in the teams helped you to perform better?</td></tr>
</table>

Communication

How is it assessed?	What is assessed?
Report	The clarity, structure and professionalism
Presentation	
Demonstration	
Individual Reflection (along with your report) – 300 words	

Self-Learning

How is it assessed?	What is assessed?
Report	
Presentation	
Demonstration	
Individual Reflection (along with your report) – 300 words	Answer the question: • What skills and knowledge did you acquire in the project? How?

Self-Learning: The Professional Development Cycle



Using the professional development cycle

Self-Learning: How to Record?

Plan		Do	Record	Review	
What ?	How?	When?		Evaluate	Next Steps

Self-Learning: What counts?



Training Courses



Work Experience



Academic Study



Volunteering



Events and Seminars



Self-Study

Agenda

1. Professional Competencies
2. Assessments
3. **Next Steps**

Next Steps

Date	Time	Theme
21/05	10.00 – 12.00	Academic writing and technical communication (Centre for Academic English)
21/05	13.00 – 15.00	Teamwork and Career Planning (Jess Popplewell, Careers Service / PR)
21/05	15.00 – 17.00	Ethics in engineering (Prof Jeremy Pitt, EEE)
08/06	10.00 – 12.00	Report formative feedback session (Centre for Academic English / EEE Staff)

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Thank you